

Request and Conquer: Exposing Cross-Origin Resource Size

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Abstract:

Numerous initiatives are encouraging website owners to enable and enforce TLS encryption for the communication between the server and their users. Although this encryption, when configured properly, completely prevents adversaries from disclosing the content of the traffic, certain features are not concealed, most notably the size of messages. As modern-day web applications tend to provide users with a view that is tailored to the information they entrust these web services with, it is clear that knowing the size of specific resources, an adversary can easily uncover personal and sensitive information.

In this paper, we explore various techniques that can be employed to reveal the size of resources. As a result of this in-depth analysis, we discover several design flaws in the storage mechanisms of

browsers, which allows an adversary to expose the exact size of any resource in mere seconds. Furthermore, we report on a novel size-exposing technique against Wi-Fi networks. We evaluate the severity of our attacks, and show their worrying consequences in multiple real-world attack scenarios. Furthermore, we propose an improved design for browser storage, and explore other viable solutions that can thwart size-exposing attacks.

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